Deep learning on audio and EEG: decoding speech acoustics from the brain

PhD project • SOUND Lab, University of Cambridge • ongoing • Python, PyTorch

Linked: SPiN 2026 poster - reconstructing sound from EEG with LLMs • WCA 2026 poster - lexical surprise / codec EEG

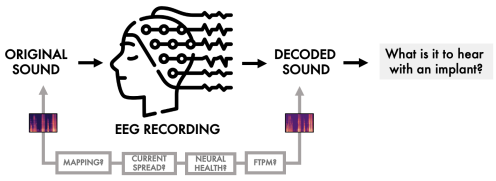
What acoustic information can be recovered from the brain of typical-hearing listeners, hearing-aid users and cochlear-implant users?

Reconstructing the speech spectrogram across hearing groups (paper in preparation)

Convolutional and attention networks (PyTorch) decode the 64-band speech spectrogram from 64-channel EEG in 81 listeners - 28 typical-hearing, 29 hearing-aid and 24 cochlear-implant users - on held-out trials, with every measure computed against a shuffled-EEG floor through identical arithmetic. Reconstructed windows are retrieved well above chance in all three groups (25%, 23% and 18%, chance 5%), lower in cochlear-implant users. In competing-talker trials, the attended talker is reconstructed selectively in every group, with weaker selectivity in implant users. The frequency centroid of the decodable evidence differs between groups, shifting lower in hearing-aid users and higher in cochlear-implant users (mixed model, $p = 0.006$; permutation across listeners, $p = 0.011$). Spectral descriptors (tilt, centroid, flux, F0) remain decodable after the broadband envelope is regressed out, and a two-stage route through interpretable descriptors separates what the acoustic representation costs from what the brain costs.

Sound reconstruction with LLM architectures

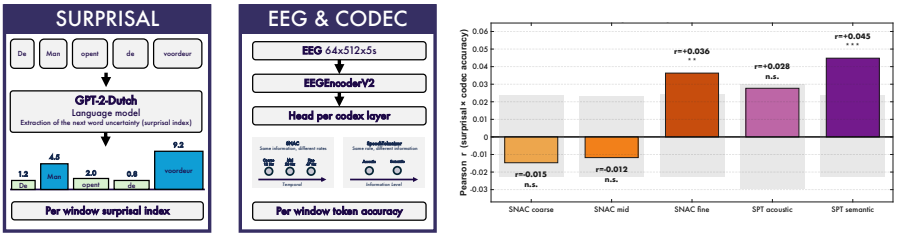
Existing decoders (linear envelope tracking, CNN/RNN spectrogram models) recover only specific acoustic components. I built a framework on large-language-model architectures - NeuroLM, a GPT-2-based model pre-trained on large-scale EEG - with sound tokenisation, to attempt full sound reconstruction from brain activity.



The reconstruction paradigm and its link to the individual factors studied.

Lexical surprise / codec EEG

Using an audio codec over 64-channel EEG and word-level surprisal from a language model, this study asks whether lexical surprise selectively amplifies phonetic versus semantic neural codes. Only the codec streams carrying lexical information correlated with the surprisal index.



Stages of the lexical-surprise study.